

An Improve Scheme of Segmenting Colour Food Image by Robust Algorithm

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Abstract

Segmentation method in digital image format is required to assist the detection of food features that could point to different pathologies. In this paper, a segmenting method is present to obtain an accurate binary map. First, the simple segmentation is separated a rough representation from the contrast image. Secondly, the boundary searching process is presented and ascertains the extent of food. Finally, some areas inside the object that was not detected are added into the segmentation. Simulation result shows that the proposed approach fits the kind food with native holes in side and is superior to others algorithm.

Keyword: *segmentation, thresholding, edge-based segmentation, region-based segmentation*

1. Introduction

The quality assurance methods used in the food industry have traditionally involved human visual inspection. Such methods are tedious, laborious, time-consuming and inconsistent. During the past decade, computers have been introduced widely for automation of the quality assurance task. In general, a machine vision sensor is used to measure some aspect of the result manufacturing process (e.g. shape, size, texture, location) that is indicative of the accuracy, efficiency or quality of the process.

Recognition of useful object feature from image data is a complex task involving a series of steps: image acquisition, preprocessing, segmentation, representation and description. As the name, implies, image segmentation refers to the process of segmenting or partitioning a composite image into component parts or objects. Autonomous segmentation is one of the most difficult tasks in image processing [1].

Segmentation is to divide an image into regions that have a strong correlation with objects or areas of interest. Generally, in the food image segmentation, there are only two regions: the foreground (food image itself) and the background. Thus, the result is a binary image, where the pixels may take only one of the values '1' or '0'. It can be achieved by three techniques: thresholding, edge-based segmentation and region-based segmentation [2-4]. There are a number of segmentation algorithms proposed for food products [5-11]. Recently, Sun and Du [12] developed an algorithm for segmenting complex images of many types of foods including pizza, apple, pork and potato. In their approach, the food image itself is segmented into different sub-regions.

Mery and Pedreschi [13] proposed a robust algorithm for segmenting food images from their backgrounds using colour image. Their scheme involves three steps: (1) computation of a high contrast grey value image from an optimal linear combination of the RGB colour components; (2) estimation of a global threshold using a statistical approach; and (3) a morphological operation in order to fill the possible holes presented in the segmented binary image. However, some kind foods have small space inside the object naturally. The morphological operators can not concern about that characteristic and regard the native holes as foreground. If a few of holes are inside the object, the scheme is filled them. In this paper, we provide a scheme to improve the drawback to adapt the kind of foods. This paper organizes as follows. In section 2, the improve scheme is described and experimental results for the proposed approach are presented in section 3. Finally, section 4 ends the paper with some conclusions.

2. Proposed Method

Fig. 1 demonstrates the step and data flow chart. There are five parts in this method including converting original image to contrast image,

deciding the global threshold, labeling the object contour, first segmenting and filling the object. The input of our approach is a RGB image and the output is a binary map that contains two values '0' and '1'.

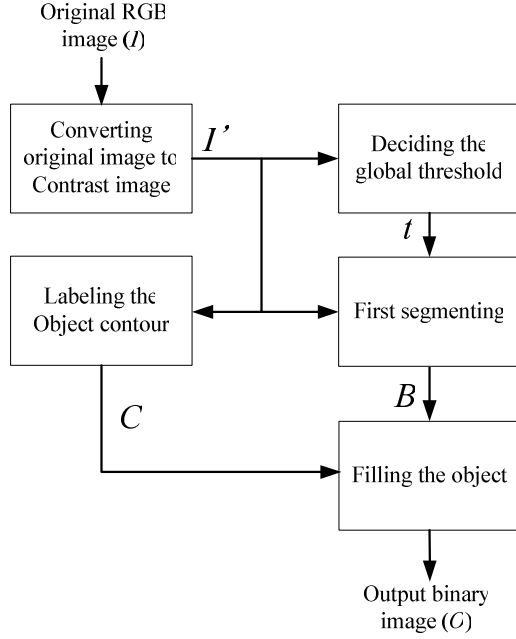


Fig. 1 Flow chart

A RGB image that is stored in three matrices, R , G , and B respectively should convert into a contrast intensity image. There are several colour space transformations [14] that attempt to obtain a better representation of the colour. The linear form is formulated as:

$$I(x, y) = k_r R(x, y) + k_g G(x, y) + k_b B(x, y) \quad , \quad x = 1, 2, \dots, M \quad \text{and} \quad y = 1, 2, \dots, N \quad (1)$$

$I(x, y)$ is the transformed grey value of the (x, y) pixel. The corresponding intensity values for a (x, y) pixel are denoted as $R(x, y)$, $G(x, y)$, and $B(x, y)$ and M, N are respectively the size of the files and columns of the original image. (k_r, k_g, k_b) are the weights that ponder the RGB components. A normalized monochrome image is computed as:

$$I'(x, y) = \frac{I(x, y) - I_{\min}}{I_{\max} - I_{\min}} \quad (2)$$

The high contrast image means a representation of image with low homogeneity that will have a high variance. The best combination (k_r, k_g, k_b) in (1) that maximizes the variance of (2):

$$\sigma_{I'}^2(k_r, k_g, k_b) \longrightarrow \max \quad (3)$$

It follows from what has been said that the high

contrast monochrome image is computed from a RGB image. The example is demonstrated in section 3.

A first separation between foreground and background can be performed estimating a global threshold t . A binary image B that is an outcome of first segmenting process is defined as:

$$B(x, y) = \begin{cases} 0 & , \text{if } I'(x, y) \leq t \\ 1 & , \text{if } I'(x, y) > t \end{cases} \quad (4)$$

$B(x, y)$ retains two classes of pixels, where '1' means foreground and '0' background. A best threshold t separates the two modes, background and foreground. The threshold is obtained by ensuring a small variation of the gray values in each class or a large variation of the gray values in the image [15].

Binary image B is the firstly segmenting by previous work. There is some area that is not detected inside the object. However, some undetected holes belong to object and some are not. The following two steps would separate the object from those undetected holes. First, edge of the contour is defined. Finally, object holes are recognized and filled value '1' in B .

Generally speaking, contour of object is the pixel with large variation. For the reason, the standard deviation and variance are performed for the step. The standard deviation and variance is preformed in 3x3 neighborhoods around the corresponding pixels in the contrast image I' shown in fig.2.

$f(x-1, y-1)$	$f(x-1, y)$	$f(x-1, y+1)$
$f(x, y-1)$	$f(x, y)$	$f(x, y+1)$
$f(x+1, y-1)$	$f(x+1, y)$	$f(x+1, y+1)$

Fig. 2 Variance and standard deviation of 3x3 neighborhoods

Equation (5) formulates the equation standard deviation, where f is the contrast image from foregoing step. Variation is simply formation as V^2 .

$$V(x, y) = \sqrt{\frac{1}{8}(f(x, y) - \bar{f})^2} \quad (5)$$

, where $\bar{f} = \frac{1}{9} \sum_{i=-1}^1 \sum_{j=-1}^1 f(i, j)$

The boundary of object is classified by equation (6). Large variation pixels that contain the variance in (5) probably regard as boundary. If a pixel belongs to boundary set, the pixel in C is significant and set to '1'.

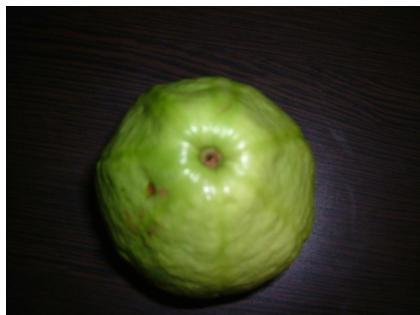
$$\begin{cases} (x, y) \in \text{boundary set} & \text{if } V^2(x, y) > m \\ (x, y) \in \text{smooth set} & \text{if } V^2(x, y) \leq m \end{cases} \quad (6)$$

, where m is the mean value of V . Some holes that belong to food body are not detected through the preceding steps. It is necessary to find out the location of those pixels and fill them up. The stuffing approach copes with each pixel row by row until all rows processed. For each row in C , the start and next points are scanned beforehand. The start point is the right hand side pixel that located on the last pixel of first significant cluster. The pixel situated at the left hand side pixel that is the first pixel of the next significant cluster is end point. If any significant pixel of B located on the segment that is from start point to end point, every insignificant pixel in C between the two points is part of object. Next, the pair of start and end point is looking for and searching process repeat until all the rows in C are scanned.

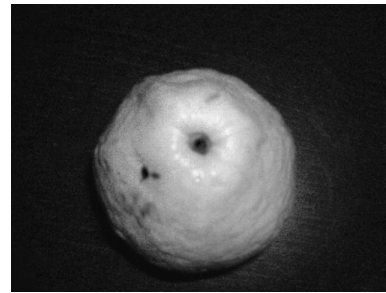
3. Simulation Result

In this section, simulation results are presented and compare with Mery's result in the same situation. The result shows our proposed scheme suit to segment the food object with irregular gap. Our segmenting result is superior to Mery and obtains a object binary map precisely. There are two examples shown in Fig.3 and Fig. 4.

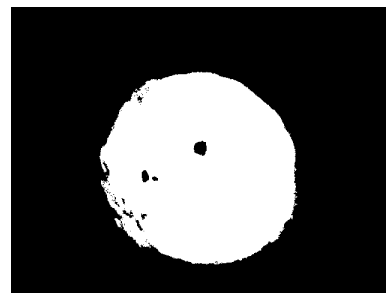
Fig.3 (a) is a RGB image that contains a guava with some defects. The original image is converted into a contrast image that composes of lots large variation areas shown in Fig. 3(b). According the global threshold, first segment map is determined and some areas that are block hole in side the object are not decided. In order to figure out whether the holes are in side, object contour should be detected in Fig. 3(d). After filling process, a binary map is obtained shown in Fig. 3(e). Our result is similar to Mery's illustrated in Fig. 5(a) because the image is only one object without naturally space.



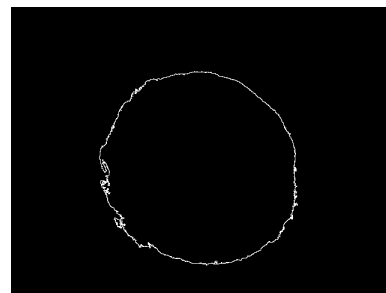
(a)



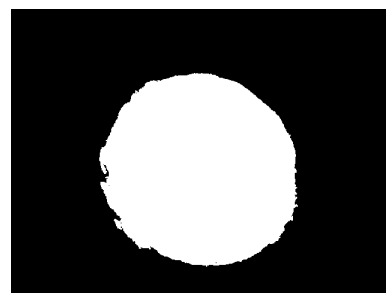
(b)



(c)



(d)



(e)

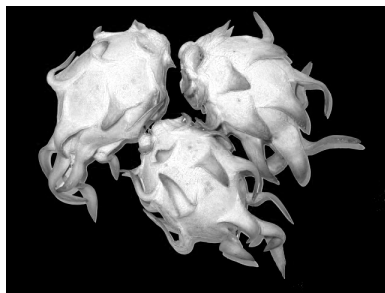
Fig 3. Guava image (a) RGB image (b) High contrast image (c) First segmenting (d)Boundary of object (e) Segmenting result

Fig. 4 (a) is a RGB image that composes of three Night-blooming Gereuses. Their leaves are

interleaving and become closed region. Most algorithms would regard these regions as object. Our method can overcome that situation and make out whether the object. The segmenting steps of Night-blooming Gereus image are shown in Fig. 4(b)-(e). Mery's result was shown in Fig. 5(b) that all the closed regions are regard as object and fill them out.



(a)



(b)



(c)

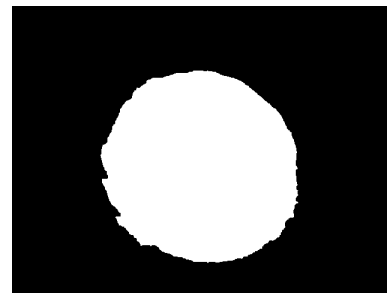


(d)



(e)

Fig 4 Night-blooming Gereus image
(a) RGB image (b) High contrast image (c) First segmenting (d)Boundary of object (e) Segmenting result



(a)



(b)

Fig 5 Simulation result of Mery's algorithm
(a) Guava image
(b) Night-blooming Gereus image

4. Conclusions

In this paper, an automatic segmentation approach is proposed to segment the accuracy object of food from background. If a few of holes are inside the object, the scheme is filled them. If holes interleave on several objects and become closed region, the scheme can overcome that situation and make out whether the object.—We provide a scheme to improve the drawback to adapt the kind of foods. Experimental results

demonstrate that our approach performs relatively better than Mery's approach. However, there are still the several aspects to explore. The blight of the fruit should be detection and the quality of food is determined. More texture of food needs to recognize in the future work.

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